OFFICIAL USER GUIDE

ADSI Inverter Dashboard

Professional plant monitoring, reporting, forecast review, and controlled remote coordination for utility-scale solar inverter fleets.

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1 Overview & Purpose

ADSI Inverter Dashboard provides centralized operational visibility for a utility-scale solar inverter fleet. The application consolidates live status, interval energy, alarms, audit activity, and formal reporting in one workstation interface.

Live Monitoring

Real-time inverter and node-level status with supervised control actions.

Analytics & Forecasting

Daily analytics with interval charts and day-ahead forecast comparison.

Alarm Management

Episode tracking with severity, duration, and acknowledgement state.

Reporting & Export

Daily performance reports and structured exports in Excel or CSV.

Remote Operations

Gateway-linked remote viewing and write control from authorized workstations.

Cloud Backup

Scheduled cloud backup with OneDrive or Google Drive integration.

2 System Requirements

REQUIREMENT	DETAILS
Operating System	Windows 10 / 11 (64-bit)
Display	1920 × 1080 minimum; scales up at 2200px+ and 3000px+
Administrator Rights	Required — the installed app requests elevation automatically
Network	LAN access to inverter fleet (Gateway mode) or secure network path to gateway (Remote mode)
Disk Space	Sufficient for database, archive, and export storage

3 Getting Started

Installation

1. Run the installer (`Inverter-Dashboard-Setup-x.x.x.exe`).
2. Accept the Windows User Account Control prompt to allow administrator elevation.
3. Choose the installation directory (default is recommended).
4. Complete the wizard and launch the application.

First Sign-In

FIELD	DEFAULT VALUE
Username	`admin`
Password	`1234`

TIP

Change your credentials after first login using the **Update Username and Password** link on the sign-in screen. You will need the Authorization Key.

Credential Management

ACTION	DESCRIPTION
Update Username and Password	Change credentials using the Authorization Key
Recover with Authorization Key	Restore default credentials (`admin` / `1234`)
Remember credentials	Persists login on the workstation

After successful sign-in, the application maximizes to full screen and begins service initialization.

4 Interface Layout

Header Bar

The fixed header is the primary global status strip, always visible at the top.

ELEMENT	DESCRIPTION
Plant Logo / Title	Identifies the plant and dashboard instance
`TOTAL POWER OUTPUT (Pac)`	Combined AC power from all online inverters in real time (kW)
`TOTAL ENERGY GENERATED`	Today's cumulative plant energy (MWh)
Clock & Date	Local workstation time, updates every second
Theme Toggle	Switch between **Dark**, **Light**, and **Classic** themes
Alarm Sound	Mute or unmute alarm audio notifications
Connection Indicator	Green dot when live data is streaming; red when disconnected
Menu Button	Toggle the side navigation panel

Side Navigation

The navigation panel provides access to all main pages:

PAGE	PURPOSE
Inverters	Live inverter status, node-level control, and real-time output monitoring
Analytics	Daily energy analytics with interval charts and day-ahead forecast comparison
Forecast	Dedicated forecast workspace for day-ahead generation and Solcast provider management
Alarms	Alarm episode history with severity, duration, and acknowledgement tracking
Energy	5-minute interval energy records by inverter for the selected date
Audit	Operator command history with timestamps, scope, and results
Report	Daily performance report with energy, output, availability, and alarm counts per inverter

PAGE	PURPOSE
Export	Export operational data, alarms, reports, and forecasts to Excel or CSV
Settings	Plant configuration, connectivity, license, updates, and cloud backup management

About Section

The bottom of the sidebar shows application and license information including installed version, license state, update state, and data directory. The **Open User Guide** button launches this guide.

Visual Themes

Three themes are available. Theme selection persists between restarts.

THEME	DESCRIPTION
Dark	Default theme with dark backgrounds and high-contrast text
Light	Light backgrounds for well-lit environments
Classic	Alternative dark palette with warmer tones

5 Inverters Page

The Inverters page is the primary live operations page for monitoring the full inverter fleet, reviewing node-level status, and sending control commands.

Toolbar Controls

CONTROL	FUNCTION
Inverter Filter	Show all inverters, or filter by Online, Offline, or Alarmed status
Layout	Change the grid column layout (Auto, 2–7 columns)
Show Cap / Hide Cap	Toggle the plant output cap panel

Fleet Status Chips

Quick counts displayed on the toolbar:

CHIP	MEANING
Inverters	Active / Total inverter count
Nodes	Active / Total node count
Online	Inverters currently producing power
Alarmed	Inverters with active unacknowledged alarms
Offline	Inverters not responding

PAC Output Band Legend

Node output level indicators based on percentage of rated node capacity:

🟢 ≥90% High 🟡 >70% Moderate 🟠 >40% Mild 🔴 ≤40% Low 🔴 Alarm

Inverter Card Contents

Each inverter card displays:

- **INVERTER XX** — title with current state badge
- **Start / Stop buttons** — inverter-wide control (icon-only in 7-column layout)

- **Pdc / Pac summary** — combined DC and AC power
- **Node table** — per-node Alarm, Pdc, Pac, Last Seen, and Control columns

Inverter Detail Panel

Clicking any inverter card opens a detail panel showing:

- Live inverter statistics (Today Energy, Peak Pac, Avg Pac, Uptime)
- Today's alarm activity
- Last 7 days performance summary

Node Controls

ACTION TYPE	SCOPE	AUTHORIZATION
Node button	Single node	None (confirmation only)
Card Start / Stop	Entire inverter (all nodes)	None (confirmation only)
Bulk START / STOP SELECTED	Selected inverter set	Authorization key required

Bulk Inverter Commands

FIELD	FUNCTION
Inverter Numbers / Ranges	Enter values like `1-13, 16, 18, 23-27`
All Inverters	Auto-fills the full valid inverter range
Clear	Clears the current selection
START SELECTED	Starts configured nodes in selected inverters
STOP SELECTED	Stops configured nodes in selected inverters

IMPORTANT

Bulk control actions (**START SELECTED** / **STOP SELECTED**) require a separate authorization key from authorized personnel. Isolated inverters are skipped automatically.

6 Plant Output Auto Capping

Automatically regulates total plant output within a defined MW band by sequentially stopping and starting whole inverters. Operators set upper and lower limits, exempt specific units, and preview the control plan before activation.

Cap Configuration

FIELD	FUNCTION
Upper Limit (MW)	Upper plant MW threshold that triggers automatic stop decisions
Lower Limit (MW)	Lower threshold for restarting eligible stopped non-exempt inverters
Sequence	Inverter selection mode: Ascending, Descending, or Exemption
Exempted Inverter Numbers	Comma-separated inverter numbers excluded from automatic stop selection
Cooldown (s)	Settling time between each automatic action (default: 30)

Cap Actions

ACTION	FUNCTION
Preview Plan	Simulates next stop or restart decision using live plant state
Enable Cap	Enables gateway-side plant output capping (requires authorization)
Disable Monitoring	Stops automation without restarting controller-owned inverters
Release Controlled Inverters	Restarts controller-owned inverters sequentially and ends the session

Cap Status Panel

The status panel reports live controller state through a set of labeled metrics:

METRIC	MEANING
Status	Controller state: Idle, Monitoring, Stopping, Starting, Paused, or Fault
Reason	Human-readable explanation of the current state or pending action

METRIC	MEANING
Last Action	Most recent automatic stop or restart, with inverter number and timestamp
Cooldown	Remaining settling time after the most recent controller action
Curtailed	Total MW removed by controller-owned stops (sum of Pac at each stop time)
Controllable	Number of inverters eligible for stop selection
Pending	In-flight controller action, if any
Exempted	Inverter numbers excluded from automatic stop selection

Controlled Inverters Table

When the controller owns one or more stopped inverters, a detailed table appears inside the cap panel:

COLUMN	MEANING
Inverter	Inverter number stopped by the controller
Stopped At	Time the controller issued the stop command
Duration	Elapsed time since the stop (e.g. 12m 35s or 2h 05m)
Pac Removed (kW)	AC power output at the moment of stop
Nodes	Enabled node count at the time of stop
Rated kW	Node-adjusted rated inverter capacity
Depend. kW	Node-adjusted dependable inverter capacity

Cap-Stopped Inverter Card Indicators

Inverters stopped by the plant cap controller are visually distinct in the inverter grid:

- The card badge changes from **OFFLINE** to **CAP STOPPED** (blue)
- A stoppage timestamp appears below the badge showing when the controller stopped the inverter
- The card border and icon shift to a blue accent instead of the dimmed gray used for regular offline inverters
- The card stays at full opacity, unlike ordinary offline cards which are dimmed

These indicators appear on all three themes (dark, light, classic) and clear automatically when the inverter is released or the cap session ends.

Cap Plan Preview

The preview table shows each candidate inverter in sequence order with its node count, rated and dependable capacity, estimated step kW, projected plant MW after the step, and the planner's decision reason. The selected candidate (next action) is highlighted.

IMPORTANT

- Current builds use whole-inverter sequential control only
- Very narrow MW bands may not settle cleanly with whole-inverter steps
- While plant cap is enabled, all non-exempted inverters stay under controller authority
- Manual start/stop for non-exempted inverters is blocked until plant cap is disabled or the inverter is exempted
- If a manual write bypasses the API guard and reaches a controller-owned inverter, the cap controller pauses as a safety fallback and requires re-enable after review
- In Remote mode, cap actions are proxied to the gateway workstation
- All cap controller stop and start actions are recorded in the Audit page with scope **PLANT-CAP**

7 Analytics Page

Interval-based review of production with day-ahead forecast comparison.

Controls

CONTROL	FUNCTION
Date	Select the day to analyze
Interval	Chart grouping: 5 min, 15 min, 30 min, or 1 hour
Load View	Loads analytics for the selected date and interval

Summary Metrics

METRIC	MEANING
Interval	Selected time bucket size
Total Energy	Plant total for the day (MWh)
Peak Plant Output	Highest combined AC power (kW)
Reporting Inverters	Count of inverters with data
Latest Interval	Timestamp of last recorded interval

Selected Date Summary

ITEM	MEANING
Actual MWh	Authoritative total daily energy — updates automatically when today's date is selected
Day-ahead MWh	Forecasted total daily energy
Variance MWh	Difference between actual and forecast
Peak Interval	Highest interval energy with timestamp

Live updates: When today's date is selected, the summary card and interval charts refresh automatically on each server push — no manual reload needed. For past dates, press **Load View**.

Day-Ahead Generator

Available on the gateway workstation only. Enter a Days value and click Generate to produce the day-ahead forecast. Export is handled on the dedicated Export page.

7-Day Weather Outlook

Daily weather context: sky condition, temperature range, rainfall, cloud percentage, wind speed, and solar irradiance. Serves stale cached data when the weather API is unavailable.

8 Forecast Page

Dedicated workspace for forecast configuration, Solcast validation, and toolkit preview.

Forecast Source Options

SOURCE	USE
Local ML (Current)	Standard local ML-trained forecasting source
Solcast	External forecast source for validation or alternate use

Solcast Access Modes

MODE	USE
Toolkit Login	Reads the Solcast toolkit chart feed using account sign-in
API Key	Uses formal Solcast API credentials and resource ID

Solcast Toolkit Settings

Instead of entering a full URL, operators configure structured fields. The application constructs the Solcast toolkit URL internally.

FIELD	DESCRIPTION
Plant Resource ID	Your plant's unique Solcast site identifier
Forecast Days	Number of days to forecast (1–7, default: 2)
Resolution	Forecast interval: PT5M, PT10M, PT15M, PT30M, PT60M (default: PT5M)
Toolkit Email	Solcast toolkit account email
Toolkit Password	Solcast toolkit account password

Toolkit Preview

CONTROL / METRIC	PURPOSE
Start Day	First day shown in preview
Days to Display	Number of days included
Chart Unit	View values as MWh or MW
Forecast Total	Total forecasted energy in selected window
Estimated Actual	Estimated actual value for comparison

Forecast Actions

BUTTON	FUNCTION
Save and Test Solcast	Saves active values, then tests the chosen Solcast mode
Refresh Preview	Reloads toolkit preview using current form values

NOTE

Day-ahead forecast generation is only available on the Gateway workstation. In Remote mode, run generation from the gateway.

DAY-AHEAD GENERATOR — ML TRAINING ENGINE

The day-ahead forecast is produced by a **trained machine-learning model** that learns your plant's unique generation profile from historical operating data. The engine runs on the gateway workstation and builds a production forecast for the next day based on weather inputs and observed performance patterns.

How Training Works

- The engine collects historical irradiance, weather, and production data to build a training dataset.
- Training samples are filtered by an **operational constraint mask** to exclude periods where the plant was not operating normally (manual stops, faults, or partial shutdowns).
- A hardened data collector rejects outliers, flat-line artifacts, and low-confidence samples to improve model quality.
- The trained model produces a full-day energy forecast at configurable intervals (5 min to 1 hour).

Plant Cap Awareness

When the **Plant Output Cap** controller stops inverters to limit plant output, the ML engine treats these periods differently from manual stops or faults:

- **Cap-dispatched stops** (logged with scope `plant-cap` in the audit log) are identified separately from manual operator stops.
- During cap-only periods (no concurrent manual stop or fault), the engine **reconstructs** what the plant would have produced at full capacity using a physics/hybrid baseline.
- This preserves high-irradiance training samples that would otherwise be lost — the model sees the reconstructed output instead of the artificially capped value.
- Only manually constrained slots are excluded from training. Cap-reconstructed slots contribute normally with near-zero residual.

Intraday Adjustment

- Throughout the day, the engine compares observed actual output against the day-ahead forecast to compute an adjustment ratio.
- **Cap-depressed observations are excluded** from this ratio so that curtailed output does not propagate as an under-forecast bias into remaining slots.
- If too few uncapped observations are available, the engine falls back to the full observation set.

Error Memory

- A rolling error correction system tracks forecast accuracy over recent days.
- All operationally constrained slots (including cap dispatch) are excluded from error correction to prevent false negative bias from curtailed actuals.

Operational Impact

SCENARIO	EFFECT ON FORECAST TRAINING
Normal operation (all inverters running)	All data used directly for training
Manual inverter stop or fault	Affected slots excluded from training
Plant cap controller stops inverters	Output reconstructed to full-capacity baseline; training preserved
Mixed (cap + manual stop together)	Slot excluded from training (treated as manually constrained)

IMPORTANT

The forecast engine depends on the `plant-cap` scope tag in the audit log to distinguish automated cap stops from manual stops. Do not modify or clear audit log records without

understanding this dependency.

Forecast Performance Monitor

The Forecast Performance Monitor provides a visual audit of the ML forecast engine: health status, accuracy trends, and a per-day comparison table for the selected look-back window. It is available on the Forecast page under the “Performance” tab.

Health Chips

CHIP	WHAT IT SHOWS
ML Training	Status of the last model training run: *Trained*, *Rejected (N consecutive)*, or *No data*
Last Run	Outcome of the most recent day-ahead generation attempt and its timestamp
Provider	Data provider used for the last run: *Local ML* or *Solcast*
Recent Quality	Aggregate quality rating over the selected window: *Good*, *Acceptable*, or *Poor*

Compare Chart

Overlays the day-ahead forecast (line) against actual generation (bars) for each date in the window. A shaded confidence band shows the forecast high–low range where available.

WAPE Chart

Plots the daily Weighted Absolute Percentage Error (WAPE) as a bar chart. Lower values indicate higher accuracy. Bar colour reflects the quality tier for that day.

History Table

COLUMN	DESCRIPTION
Date	Target date of the forecast
Provider	Data provider used (*Local ML* or *Solcast*)
Variant	Forecast variant tag (e.g. *day_ahead*)
WAPE %	Weighted Absolute Percentage Error for that day
Forecast MWh	Forecasted daily energy total

COLUMN	DESCRIPTION
Actual MWh	Observed actual energy total
Freshness	Solcast input freshness classification
Quality	Overall quality tier for that forecast run
In-Memory	Whether the forecast is held in the in-memory error-correction pool

Controls

CONTROL	FUNCTION
Day-range selector	Sets the look-back window: 7, 14, 30, 60, 90, or 180 days
Refresh	Reloads all panel data from the server

ML Backend — LightGBM

LIGHTGBM INSTALLATION & REQUIREMENTS

The forecast engine uses **LightGBM** as its primary ML backend when available (enabled by default from v2.4.40). LightGBM delivers faster training and improved accuracy on tabular solar data compared to the built-in sklearn Gradient Boosting Regressor fallback.

Installation

Install LightGBM into the Python environment used by the Forecast Service:

```
pip install lightgbm
```

If LightGBM is not installed the engine falls back automatically to sklearn GBR — no configuration change is required. Switching is fully transparent: the same training pipeline and hyper-parameters apply.

Requirements

REQUIREMENT	DETAILS
Python	3.8 or later
LightGBM package	3.x or later — `pip install lightgbm`
Visual C++ Redistributable (Windows)	Usually already present; required by the LightGBM DLL
CPU / RAM	No GPU required; standard workstation hardware is sufficient

REQUIREMENT	DETAILS
Disk	~50 MB for the LightGBM package and its DLLs

Verifying the Active Backend

- Check the **ML Training** health chip in the Forecast Performance Monitor after the next training run.
- The Forecast Service log prints `[LightGBM]` entries during model fit when LightGBM is active.
- To force the sklearn fallback (e.g. for debugging), set the environment variable `FORECAST_USE_LIGHTGBM=0` before starting the Forecast Service.

NOTE

PyInstaller builds bundle LightGBM automatically if it is installed on the build machine. If the build machine does not have LightGBM, the packaged EXE will use the sklearn fallback at runtime.

9 Alarms Page

Formal alarm review and acknowledgement workspace.

Controls

CONTROL	FUNCTION
Inverter	Filter by specific inverter or all
Date	Select review date
Load Records	Query alarms for the selected scope
Min Duration	Filter out alarms shorter than N seconds
Acknowledge All	Acknowledge all active alarms in scope

Alarm Table Columns

COLUMN	MEANING
Alarm Time	Timestamp when alarm was first raised
Inverter	Inverter identifier
Node	Node number within inverter
Alarm Code	Hexadecimal alarm code from inverter register
Severity	Classification: Critical, Warning, Info
Description	Human-readable alarm description
Cleared	Time the alarm resolved (or — if still active)
Duration	Active or total duration
Status	Active or Cleared
Ack.	Acknowledged or unacknowledged

Alarm Sound Behavior

- Sound triggers only for unacknowledged active alarms lasting at least **5 seconds**
- Very short alarm blips are ignored to prevent false noise
- If a node already has an active alarm and the alarm value changes, the same episode continues — no new sound is triggered
- Use the header mute button to silence alarms without disabling detection

10 Energy Page

5-minute interval energy records by inverter for detailed production verification.

KPI Summary Tiles

TILE	MEANING
Date Total	Total MWh for selected date across all inverters
Average per Interval	Mean MWh per 5-minute bucket
Peak Interval	Highest single interval value with timestamp
Reporting Inverters	Count of unique inverters with data
Latest Interval End	End time of the most recent recorded interval

Live updates: When today's date is selected, all KPI tiles update automatically on each server push — no manual reload needed. For past dates, press **Load Records**.

Energy Table Columns

COLUMN	MEANING
Date	Calendar date of interval
Interval End	Timestamp when 5-min bucket closed
Inverter	Inverter identifier
Interval Energy (MWh)	Energy produced during that interval

11 Audit Page

Operator command accountability and traceability records.

Audit Table Columns

COLUMN	MEANING
Date/Time	Command timestamp
Operator	Operator name recorded at execution
Inverter	Affected inverter
Node	Affected node or aggregated scope (ALL if the command targeted all nodes)
Action	START or STOP
Scope	SINGLE, INVERTER, SELECTED, ALL, or PLANT-CAP
Result	OK or ERROR
IP	Source workstation or inverter IP address
Reason	Controller decision reason for automatic actions (plant-cap scope); blank for manual commands

All columns support sorting. An inline filter row enables targeted review by operator, action, scope, result, or IP address.

Plant-Cap Scope Indicator

Audit entries generated by the Plant Output Cap controller display a **PLANT-CAP** badge in the Scope column with a blue highlight on the row. The **Reason** column shows the controller's decision reason (e.g. "Keeps projected plant output above the lower limit."), making it straightforward to distinguish automatic cap actions from manual operator commands.

12 Report Page

Daily inverter-by-inverter performance summary for formal reporting and management handoff.

Report Table Columns

COLUMN	MEANING
Inverter	Inverter identifier
Energy (MWh)	Daily energy total
Peak Pac (kW)	Highest AC output for the day
Avg Pac (kW)	Average AC output during the window
Uptime (h)	Operating hours
Alarms	Alarm episode count
Availability (%)	Uptime as percentage of the inverter's dynamic production window (see below)
Performance (%)	Ratio of actual to expected output

Availability Calculation

HOW IT WORKS

Availability is calculated **per inverter** using a **dynamic production window** rather than a fixed solar window:

- **Window Start** — the timestamp of the first non-zero PAC reading for that inverter, at or after 5:00 AM.
- **Window End** — the timestamp of the last interval end (zero-PAC transition) for that inverter, at or before 6:00 PM (or current time if today).
- **Formula** — `Availability = (Online Uptime / Dynamic Window) × 100%`

This ensures inverters are not penalized for the gap between the fixed 5 AM boundary and their actual generation start, or between their last generation and 6 PM.

PLANT AVAILABILITY

The **TOTAL** row averages availability across **all configured inverters**, including inactive ones. Inverters that did not produce any energy contribute 0% to the plant average, reflecting true fleet-level availability.

Reports can be exported in Excel or CSV format.

13 Export Page

Dedicated export workspace with six structured export packages.

Alarm History Export

Alarm records for selected date including code, severity, duration, and acknowledgement state.

Energy Summary Export

Daily operational energy summaries by inverter and reporting node with calculated totals.

Day-Ahead Forecast Export

Forecast-versus-actual comparison. Source: Analytics (ML) or Solcast. Format: Standard or Average Table.

Operational Data Export

Interval-sampled inverter operating data for analysis and troubleshooting.

Operator Audit Export

Operator command history with timestamp, scope, result, and affected equipment.

Daily Performance Report

Multi-day inverter performance summary with energy, availability, and alarm counts.

Common Export Behavior

- Exports are written to the configured export folder
- Format options: Excel (.xlsx) and CSV
- XLSX exports include fitted column widths, colored headers, bordered cells, and highlighted total rows
- Cancel button is available during active exports
- Result text shows filename, file size, and completion timestamp

14 Settings

The administrative center organized as section-based configuration.

Global Actions

ACTION	FUNCTION
Save Settings	Saves the active configuration
Export Settings	Downloads a configuration file (JSON)
Import Settings	Loads a saved configuration file
Export Folder...	Select the export destination path
Open Folder	Opens the current export path in Explorer
IP Configuration	Opens the network configuration window
Restore Defaults	Resets settings with confirmation

Plant Configuration

FIELD	DEFAULT	USE
Plant Name	ADSI Plant	Display and document naming
Operator Name	OPERATOR	Recorded operator identity in audit entries
Inverter Count	27	Total inverters configured
Nodes/Inverter	4	Reporting nodes per inverter
Plant Latitude	—	Geographic latitude for weather context
Plant Longitude	—	Geographic longitude for weather context

Data & Polling

FIELD	USE
Data API URL	Read-side service endpoint
Write API URL	Command/write service endpoint
Export Folder	Destination for generated files
Retention Window (days)	Days kept in main database before archival
Modbus Timeout (s)	Max wait time for read response
Reconnect Delay (s)	Delay before read retry after reconnect
Read Spacing (s)	Pause between register groups

Plant Output Cap Defaults

These settings define the default values loaded into the live Plant Output Cap panel. The settings page also shows a planner summary for selection mode, band gap, controllable inverter count, and smallest step.

FIELD	USE
Upper Limit (MW)	Default upper plant MW cap threshold
Lower Limit (MW)	Default lower plant MW threshold
Sequence	Default inverter selection mode
Exempted Inverter Numbers	Default comma-separated exclusion list
Cooldown (s)	Default settling delay after each controller action

Connectivity & Sync

FIELD	USE
Operation Mode	Gateway or Remote
Remote Gateway URL	URL of the gateway workstation
Remote API Token	Shared API token for remote access
Tailscale Device Hint	Optional identifier for secure-network status

Connectivity actions include: Test Remote Gateway, Check Tailscale, Refresh, and Refresh Standby DB.

Runtime Health

Displays real-time application health: CPU, memory, uptime, polling cycles, fetch errors, rows persisted, and connected clients.

License

FIELD	PURPOSE
Status	Current license state
Source	Where the license was loaded from
Expiry	License expiry date
Remaining Days	Days remaining on current term
Upload Replacement	Apply a new license file
Refresh License	Reload license state

App Updates

ACTION	FUNCTION
Check for Updates	Query the update channel for newer builds
Download Update	Download the update package when available
Restart & Install	Orderly shutdown then installer handoff

IMPORTANT

Do not power off the workstation or relaunch the app manually while the restart/install handoff is in progress. After restart, wait for the first local poll cycle before relying on fresh data.

15 Operating Modes

MODE	PURPOSE	MAIN BEHAVIOR
Gateway	Local plant-connected workstation	Polls and persists plant data locally; authoritative source of truth
Remote	Gateway-linked viewer workstation	Streams live data from gateway; proxies historical access; no local persistence

Gateway Mode

- Polls the plant directly and persists data locally
- Is the authoritative source of truth for live data, reports, forecasts, and replication
- Is the only mode that can generate day-ahead and intraday-adjusted forecasts

Remote Mode

- Displays live gateway data in-memory only — no local DB persistence from the live stream
- Historical views, reports, analytics, and exports are served from the gateway via proxy
- Inverter on/off write control stays enabled via gateway proxy
- Forecast generation is disabled
- When gateway is unavailable, historical pages show “Gateway unavailable”

Standby Database Refresh

The **Refresh Standby DB** action downloads the gateway main database for local standby use. The staged database is not activated until a restart.

BEST PRACTICE

Use **Refresh Standby DB** before switching a remote workstation back to Gateway mode to ensure fresh local history is available.

Gateway Link Health States

STATE	MEANING
Connected	Live data streaming normally

STATE	MEANING
Degraded	Data arriving with elevated latency
Stale	Last snapshot shown while fresh data is temporarily unavailable
Disconnected	No active connection to the gateway
Auth Error	API token mismatch or expired
Config Error	Gateway URL or configuration is invalid

Connectivity Options

Remote-to-gateway connectivity can use any of these methods:

- **Tailscale** — Zero-config mesh VPN; recommended for simplicity and security
- **Traditional VPN** — Site-to-site or client VPN (OpenVPN, WireGuard, IPsec)
- **Port Forwarding** — Direct port exposure through the router/firewall (requires HTTPS and strong tokens)
- **Cloud Relay / Tunnel** — Services like Cloudflare Tunnel or ngrok for NAT traversal
- **LAN / Same-Network** — Direct IP access when both machines share a network

16 Operator Messaging

A compact gateway-to-remote note channel for operational communication between workstations.

- Access via the floating message bubble in the bottom corner
- Messages are stored on the gateway and synced to remote workstations
- Audio notification for new inbound messages (silent for self-sent echoes)
- Clear thread action with confirmation and gateway sync
- Rate limited: 10 messages per 60 seconds per machine

17 Cloud Backup & Restore

Supported Providers

PROVIDER	AUTHORIZATION
OneDrive	Microsoft OAuth via Azure Client ID
Google Drive	Google OAuth via Client ID and Client Secret
S3-compatible	Static access key pair validated against the configured endpoint, bucket, and prefix; unchanged backup data is reused across later backups

Backup Policy

SETTING	OPTIONS
Schedule	Manual only, Daily at 3:00 AM, Every 6 hours
Scope	Application data, Configuration files, Logs (optional)
Backup date tag	Date tag used for package labeling

Application data includes the active SQLite database plus forecast model bundles, forecast history context, Solcast reliability artifacts, weather cache, and forecast snapshots stored under ProgramData.

Backup Actions

ACTION	FUNCTION
Backup Now	Create and upload a backup package immediately
Refresh Cloud List	List available cloud backups
Refresh backup history	Refresh local backup activity and restore list

Restore Behavior

- Restore creates a safety backup first
- Restore overwrites the active database and configuration
- Application restart is required to complete the restore

18 Standard Operating Workflows

Daily Startup Check

1. Launch the application and complete sign-in.
2. Wait for the startup loading screen to finish before evaluating live values.
3. Confirm the license notice area is clear.
4. Check the header connection dot and clock.
5. Review **TOTAL PAC** and **TODAY MWh**.
6. Open the **Inverters** page and confirm fleet status chips.
7. If operating remotely, confirm Connectivity & Sync status in Settings.

Live Control Workflow

1. Open the **Inverters** page.
2. Select the target inverter or node.
3. Review current alarm state and freshness.
4. Send the required START or STOP action.
5. Confirm the result via status updates and audit history.

Plant Output Cap Workflow

1. Open the **Inverters** page and click **Show Cap**.
2. Set **Upper Limit** and **Lower Limit** in MW.
3. Choose the inverter **Sequence** and add exemptions if needed.
4. Click **Preview Plan** to review proposed actions.
5. Click **Enable Cap** and complete authorization.
6. Monitor status, reason, controlled count, and planner warnings.

7. Cap-stopped inverter cards show a blue **CAP STOPPED** badge with the stoppage time for at-a-glance identification.
8. Review the **Controlled Inverters** table inside the cap panel for duration, removed Pac, rated kW, and dependable kW per inverter.
9. Use **Release Controlled Inverters** to end the session, or **Disable Monitoring** to pause automation without restarting.
10. Check the **Audit** page for a full record of cap controller actions (scope: **PLANT-CAP**) with decision reasons.

Daily Performance Review

1. Open **Analytics** for interval review and day-ahead comparison.
2. Open **Energy** for interval production detail.
3. Open **Report** for the formal daily inverter summary.
4. Export the required package from **Report** or **Export**.

Remote Standby Refresh Workflow

1. Confirm the workstation is in **Remote** mode.
2. Open **Settings** → **Connectivity & Sync**.
3. Review gateway link and transfer monitor status.
4. Run **Refresh Standby DB**.
5. Allow the preflight phase to finish before expecting the heavier snapshot transfer to begin.
6. If newer local data is detected, decide whether to cancel or **Force Pull**.
7. Wait for completion and confirm success.
8. Restart the application when the refreshed DB is needed.
9. After restart, wait for the startup loading screen and first local poll cycle before relying on live totals.

Cloud Backup Workflow

1. Configure provider access and authorize.
2. Save backup settings and select scope.
3. Run **Backup Now** or rely on the configured schedule.
4. Review history and cloud file listings.

19 IP Configuration

The **IP Configuration** window manages per-inverter network and operational settings. Open it from **Settings → IP Configuration** or Ctrl + I. Access requires an auth gate key (`adsiM` or `adsiMM`, where M is the current minute). The session lasts 1 hour.

Configuration Table

Each of the 27 inverters has one row with the following fields:

COLUMN	DESCRIPTION
Inverter	Inverter number and device label (INV-01 – INV-27). Click the gear icon to open the inverter web page.
IP Address	IPv4 address of the inverter on the local network.
Polling Interval (s)	How often the gateway polls this inverter, in seconds (min 0.01, default 0.05).
Enabled Units	Which nodes (1–4) are active. Use **All** to toggle all four. An empty selection disables the inverter.
Loss %	Estimated MW transmission loss from this inverter to the substation (0–100%). Accounts for cable degradation or distance. Used exclusively by the forecast engine for substation-level accuracy; it does *not* affect live dashboard readings, energy totals, or exports.
Save	Saves the individual row. Use **Save All Changes** at the bottom to save every row at once.

Loss % and Forecasting

When a non-zero Loss % is configured, the day-ahead forecast engine adjusts historical 5-minute energy data per inverter before training. This means the ML model learns substation-level output patterns rather than raw inverter output, producing more accurate day-ahead and intraday-adjusted forecasts.

Example: if INV-15 has a 2.5% loss (degraded cable) and INV-26 has 1.0% (far from substation), the forecast engine reduces their historical energy contributions by those percentages when building training data, computing error corrections, and scoring forecast quality.

Additional Controls

- **Check Status** – scans all configured IPs for reachability and shows an online count.
- **Open Topology** – opens the visual plant topology map.

- **Theme toggle** – switches between light and dark mode for this window.

20 Keyboard Shortcuts

SHORTCUT	FUNCTION
Ctrl + T	Open the Topology window
Ctrl + I	Open the IP Configuration window
Ctrl + =	Zoom in
Ctrl + -	Zoom out
Ctrl + 0	Reset zoom
Ctrl + L	Theme toggle in auxiliary windows

Interface Tips

- Use the alarm bell panel to quickly review and **acknowledge alarms without changing pages** — each unacknowledged alarm has an inline ✓ **ACK** button; alarm toast pop-ups also include an **ACK** button
- Use the operator message bubble for shift notes and remote coordination
- Use **Open Folder** in Settings to verify export output quickly
- Hover over controls and metrics for tooltip descriptions
- Use **Check for Updates** and **Refresh License** during planned maintenance windows

21 Troubleshooting

SYMPTOM	LIKELY CAUSE	RECOMMENDED ACTION
Connection dot shows disconnected	Live link is unavailable	Check mode, gateway URL, token, and Tailscale status
Stale status appears	Last retained snapshot is being shown	Check live link health and recent gateway contact
Refresh Standby DB completes but data unchanged	Standby DB is staged, not yet active	Restart the application
Refresh Standby DB blocked with newer-local warning	Local standby copy has newer data than gateway	Cancel to preserve local data, or use Force Pull if overwrite is intentional
TODAY MWh looks old right after startup or after switching to Gateway	Local polling has not caught up yet	Run Refresh Standby DB if needed, restart, and wait for the startup loading screen plus first poll cycle
Day-ahead generation unavailable	Workstation is in Remote mode	Run generation from the gateway workstation
Plant-cap fails with “Cannot POST”	Older gateway build or wrong remote gateway target	Update gateway app and verify Remote Gateway URL
Cap band warning: limits too close	Whole-inverter step size exceeds configured deadband	Increase gap between Lower and Upper limits
Cloud restore unavailable	Provider or backup state not ready	Refresh cloud list and verify provider authorization
IP Configuration cannot be opened	Current mode is Remote or access restricted	Use the gateway workstation
Alarm sound is silent	Sound muted or system audio unavailable	Re-enable alarm sound and check workstation audio
Export fails	Path, date, format, or dataset issue	Verify export folder, input filters, and current mode

22 Quick Reference

Defaults & Constants

ITEM	VALUE
Default Username	`admin`
Default Password	`1234`
Default Plant Name	ADSI Plant
Default Operator Name	OPERATOR
Inverter Rated Capacity	997.0 kW
Inverter Dependable Capacity	917.0 kW
Nodes per Inverter	4
Node Rated Capacity	249.25 kW
Solar Window	5:00 AM – 6:00 PM (availability uses per-inverter dynamic window within these bounds)
Alarm Sound Threshold	5 seconds sustained

Page Quick Finder

NEED	BEST PAGE
Live plant status	Inverters
Interval production analysis	Analytics and Energy
Day-ahead comparison	Analytics and Export
Forecast source setup	Forecast
Alarm review and acknowledgement	Alarms
Operator action traceability	Audit

NEED	BEST PAGE
Formal daily summary	Report
File generation	Export
Mode, link, standby DB	Settings → Connectivity & Sync
License and updates	Settings → License / App Updates
Backup and restore	Settings → Cloud Backup & Restore
Network configuration	IP Configuration
Visual plant map	Topology

Data Units

ITEM	UNIT	MEANING
PAC	W or kW	Active AC output power
PDC	W or kW	DC-side input power
Energy	MWh	Produced electrical energy
Duration	sec / min / hrs	Event age, uptime, or interval

Status Terms

STATUS	MEANING
Online	Fresh data available, unit communicating normally
Offline	No current live data available
Stale	Last snapshot shown while fresh data is temporarily unavailable
Alarm	Alarm condition is active
Acknowledged	Alarm has been operator-acknowledged

ADSI Inverter Dashboard v2.4.43

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